

Week 5 Workshop Solutions

Question 1

- (a) (i) For sub-period 1, the estimated beta is 0.7442 and its standard error is 0.1076
(ii) For sub-period 2, the estimated beta is 1.0203 and its standard error is 0.1297
(b) The hypothesis is

$$\begin{aligned}H_0 &: \beta_1 = 1 \\H_1 &: \beta_1 \neq 1\end{aligned}$$

Sub-periods 1 and 2, denoted by t_1 and t_2 , respectively

$$t_1 = \frac{0.7442 - 1}{0.1076} = -2.3773, \quad t_2 = \frac{1.0203 - 1}{0.1297} = 0.157$$

The degrees of freedom are 58 and so the critical value is ± 2.0017 . In R `qt(0.975, df = 58) = 2.0017`. In sub-period 1 we reject H_0 but not in sub-period 2.

- (c) For sub-period 1, the estimated magnitude of idiosyncratic risk is 3.872% per month, which is the volatility coming from company-specific shocks such as earnings news, product issues, management changes, and so on. Since idiosyncratic risk can be reduced through diversification, it is not priced in the CAPM.

Question 2

We test if the multi-factor model reduces to CAPM:

$$\begin{aligned}H_0 &: \beta_2 = \beta_3 = \beta_4 = 0 \\H_1 &: \text{at least one is non-zero}\end{aligned}$$

Under CAPM

$$\begin{aligned}1.979 &= \sqrt{\frac{RSS_0}{346}} \\RSS_0 &= 1355.1\end{aligned}$$

and under multi-factor

$$\begin{aligned}1.933 &= \sqrt{\frac{RSS_1}{343}} \\RSS_1 &= 1281.6\end{aligned}$$

and hence the J test

$$\begin{aligned}J &= \frac{RSS_0 - RSS_1}{RSS_1 / (T - K - 1)} \\&= \frac{1355.1 - 1281.6}{1281.6 / 343} \\&= 19.67\end{aligned}$$

5% level of significance. Comparing this J test with a χ^2_3 critical value = 7.815. In R , `qchisq(0.95, df = 3) = 7.815`. We reject H_0 . We have sufficient evidence to suggest that the multi-factor model does not reduce to CAPM.